MSA-DATI07-03 · ADVANCED EDA & DATA COLLECTION

Final project brief

Collect real data from a public REST API, make its quality documented and reproducible, and hand a modeller a notebook and a report they can trust.

What you hand in

Deliverable	What “done” means
Project notebook	Your dataset’s starter notebook, every TODO completed. **Restart & Run All** passes and prints the **same fingerprint twice**.
requirements.txt	Written by the cell in Section 2; only the packages you import, with versions.
Raw data + README	`data/raw/raw.parquet` and its README (source, collection date, filters, columns).
Data-quality report	2 pages max, from `DATA_QUALITY_REPORT_TEMPLATE.md`. Section 6, “What cannot be trusted”, is the most important.
Git history	Regular commits by every member. Your history is part of the evaluation.
Oral defence	Pitch + Q&A, plus one individual question on a data-quality decision in your table.

Rules of the game

- **One dataset per group**, from the list on the next page. Several groups may share a dataset if they ask **different questions**.
- **Real data, collected by you**, through the API, with the starter notebook’s cached session. No downloading a CSV by hand.
- **Use Claude Code as much as you want for the code. The decisions are yours**: every “why” in your decision table must be something you can explain alone, without your laptop.
- **Evidence before conclusions.** A missingness mechanism (MCAR / MAR / MNAR) or an outlier decision only counts if the notebook shows the test or the rows behind it.
- **Check what you didn’t write.** Column names, API parameters and units must come from the real data (`df_raw.columns`, the overview table), not from a guess.

Key dates

Milestone	Date
Lab session: Sections 1–6 drafted, first commit pushed	[date]
Notebook, requirements.txt, raw data and report submitted	[date, time]
Oral defences	[date]

THE DATASETS

Eight APIs, one starter notebook each

All are French open data, free, and need **no API key**. Difficulty reflects the API and the data, not the question you choose. Datasets 1, 2, 4 and 5 share the same API (Opendatasoft), so the same pagination code works for all four.

#	Dataset	Source / API	Level	What you will run into
1	Paris trees *Les arbres*	opendata.paris.fr Opendatasoft	★ **Easy**	Heights and circumferences of 0 or absurd; missing species; round-number estimates. `project_01_paris_trees.ipynb`
2	Fuel prices *Prix des carburants, flux instantané v2*	data.economie.gouv.fr Opendatasoft	★ **Easy**	Stale prices; fuels missing per station; typos (1.9 vs 19); data is a snapshot. `project_02_fuel_prices.ipynb`
3	French companies *API Recherche d'Entreprises*	recherche-entreprises.api.gouv.fr	★★★ **Hard**	Nested JSON to flatten; ceased companies still listed; coded, often missing headcounts; 25 results per page. `project_03_entreprises.ipynb`
4	TGV punctuality *Régularité mensuelle par liaisons*	ressources.data.sncf.com Opendatasoft	★ **Easy**	Missing months (COVID, strikes); station names that change; cause shares not summing to 100%. `project_04_tgv_punctuality.ipynb`
5	Regional electricity *éCO2mix régional consolidé*	odre.opendatasoft.com Opendatasoft	★★ **Medium**	Daylight-saving gaps and duplicates; structurally empty sources; meaningful negative values; time zones. `project_05_eco2mix_regional.ipynb`
6	Home energy diagnoses *DPE logements existants*	data.ademe.fr data-fair API	★★★ **Hard**	200+ columns to select from; heavy missingness; impossible surfaces and consumptions; repeated diagnoses. `project_06_dpe_logements.ipynb`
7	River water temperature *Hub'Eau, température des cours d'eau*	hubeau.eaufrance.fr	★★ **Medium**	Sensor gaps; spikes from sensors out of the water; different sampling frequencies; depth limit on paging. `project_07_hubeau_river_temperature.ipynb`
8	Food products *Open Food Facts*	world.openfoodfacts.org search API v2	★★ **Medium**	Crowd-sourced missingness; values over 100 g per 100 g; duplicate barcodes; unit mix-ups; strict rate limit. `project_08_open_food_facts.ipynb`

Choosing well

- **Hard (★★★)** datasets are for groups comfortable with nested JSON or very wide tables. Pick one only if you want the challenge.
- **Run the collection cell in the first 10 minutes.** If no rows come back, tell the teacher immediately and switch dataset: don't lose the session on a broken API.
- Pick **one question** per group. Each notebook suggests three; you may write your own, as long as the data can answer it.

WHAT YOU HAVE TO DO

Six sections, one checklist

Your starter notebook already contains the parameters cell, a cached and polite HTTP session, the collection code for your API, and the raw Parquet + README step. Everything marked **TODO** is yours.

	Section	Checklist
1	**Context & provenance**	• Business question, collection date, filters applied • Unit of observation (one row = …), confirmed on the data • Known limitations: caps, snapshots, what the API does not give you
2	**Setup & parameters**	• Every path, filter, seed and threshold in the parameters cell, nowhere else • requirements.txt written and committed
3	**Collection, ingestion & schema**	• Collect, flatten, save raw Parquet + README • Select the columns you keep and **log that selection** • pandera schema: every kept column, `strict=True`, ≥ 4 domain checks, each `nullable` justified • Validate raw data with `lazy=True`; keep the failure summary for the report
4	**Data quality**	• Duplicates: exact and near-duplicates (same entity, different id or spelling) • Impossible values, per column • Missingness: name the mechanism for your 2–3 key columns **and show the test** • Outliers: triage errors vs real extremes; never a silent drop • Every decision = one `log(...)` row; decision table saved
5	**Exploration**	• 3 figures: univariate, bivariate, multivariate • Each with a title that states the finding, a caption, and a *so what* for a modeller
6	**Findings & next steps**	• 3 findings, each tied to a figure • What cannot be trusted: column, subset, why • 5 testable hypotheses: variable, direction, population, test • Clean schema passes; same fingerprint on two Restart & Run All

Suggested plan for the lab session (2h30)

Time	Work	Sections
0:00–0:20	Pick a question, run the collection, read the overview table	2, 3
0:20–0:35	Fill the provenance card	1
0:35–1:10	Choose columns, write and validate the schema. **At 1:10, good enough: move on.**	3
1:20–2:05	Duplicates, impossible values, missingness, outliers: 4–6 decision rows	4
2:05–2:30	Captioned figures, findings, “what cannot be trusted”; commit and push	5, 6

Prepare for the defence as you go: for each row of your decision table, one member should be able to answer “why this decision, and what would change if you had done the opposite?”